

## ICS 4111: Embedded Systems & IoT

### Semester Project: Deliverable 2

**Objective:** Develop prototypes of embedded devices

#### Instructions

Using the schematics from Deliverable 1, create physical and simulated prototypes based on these device architectures. Simulations will be done through [Wokwi](#):

- a. 1 ESP32S connected to 1 MQ-5, 1 DHT22 and 1 LCD (both physical and simulated models are expected)
- b. 1 ESP32S connected to 1 MQ-5 interfaced directly with another ESP32S connected to 1 DHT22 [develop EITHER a physical or simulated model – interchangeable with architecture (c)]
- c. 1 ESP32S connected to 1 DHT22 connected to 1 relay which is connected to another ESP32S connected to 1 MQ-5 [develop EITHER a physical or simulated model – interchangeable with architecture (b)]

**FYI:** *by interchangeable it means that if you develop a physical model for (b) then develop a simulation of (c) and vice versa. In the end, your team should have at least 4 prototypes: 2 physical prototypes and 2 simulated prototypes.*

**Note:** Display suitable output for all prototypes. For physical prototypes, please use resistors to avoid blowing up components!

Once done, take images of working physical prototypes, and output on the IDE and LCD.

**If by chance you experience technical prototyping issues that couldn't be resolved before the deliverable due date, ensure you clearly document the problem, all possible solutions explored and recommend for any changes to resolve the issue.**

For simulations, save each prototype as individual public projects on Wokwi and extract the link.

For submission, generate a Markdown document (.md) in your GitHub repo and include all your content for the deliverable in the document. You can refer to formatting

guidelines Markdown docs [here](#). Submit the project repo link to e-learning before the due date. Remember to include evidence of groupwork in your submission.